

StimulHeat: a Clip-On, Low-Energy and Wireless Device for Thermal Feedback when Grasping in VR

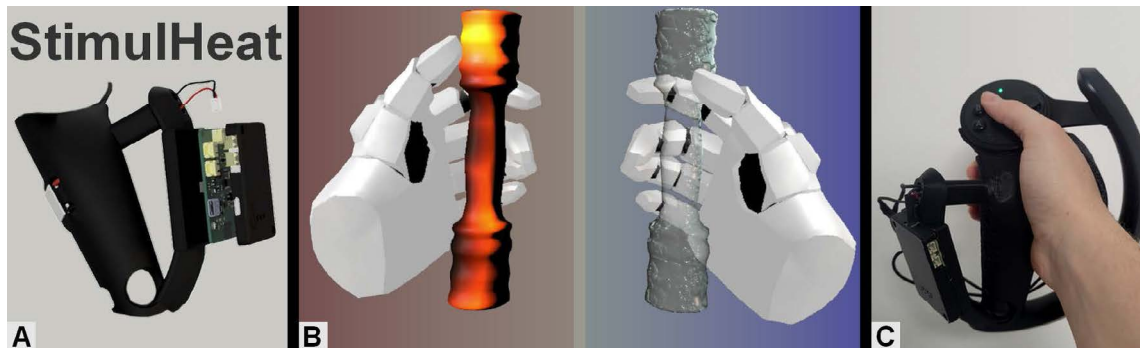
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Figure 1: StimulHeat is a non-intrusive thermal interface to feel virtual objects when grasping, no strings attached: A) CAD model of StimulHeat; B) Grasping two virtual objects with different thermal properties; C) Valve Index controller augmented with StimulHeat.

ABSTRACT

Adding thermal feedback to Virtual Reality (VR) experience enhances user immersion and the resulting feeling of presence. Our device, StimulHeat, provides a wireless, low-power and easy-to-use solution for delivering hot and cold sensations in VR, particularly when grasping objects. This demonstration highlights its integration with Valve Index Controllers, though it can be adapted to various controllers or tangible interfaces for both augmented and mixed reality experiences.

Index Terms: Virtual Reality, Haptics, Wireless, Thermal Feedback, Interactions

1 INTRODUCTION

Among possibilities of sensory stimulation in VR, thermal feedback is critical for several applications, such as firefighting training, chemical risk training, or material design. The literature contains various wearable devices for VR use that provide localized thermal feedback on specific body parts. Most are based on Peltier elements, which are controllable and can provide both cold and warm sensations. However, current integrations of Peltier have three major drawbacks for a comfortable VR usage: (1) they are energy-intensive, as such devices require large power-supplies that makes them heavy; (2) Peltier elements are controlled using Pulse-Width Modulation (PWM), which is not energy-efficient and can damage the cells overtime; (3) Proportional-Integral-Derivative (PID) controller algorithms are based on temperature value, which is not analogous to the human perception system, based on heat flow.

To address these limitations, we present StimulHeat, a novel integration of a Peltier element which is controlled via Bluetooth Low

Energy (BLE), low-power and seamlessly integrated to VR experiences. To achieve this, our device controls the Peltier element with direct current (DC) and thermal sensation is provided by regulating heat flow, rather than setting a specific temperature. Regulating heat flow is analogous to the way the human body functions, maintaining an internal body temperature of around 37°C, by balancing heat gain and heat loss [8] through vasodilation or vasoconstriction. Thus, humans do not perceive absolute temperature when touching objects, rather they perceive the resulting heat flow [3]. Material thermal effusivity provides an illustrative example of this. In a room where the ambient temperature is maintained at 20°C, objects made from different materials (e.g., wood, marble, steel, wool) will be thermally perceived differently on touch, despite the fact their measured temperature is the same. This can be attributed to the differing mechanisms of heat diffusion across different materials.

As StimulHeat mimics the human perception system by using heat flow control, we hypothesize that it will enhance the quality of haptic experience. We are currently designing a psychophysical study using StimulHeat to help validate this hypothesis and better understand its effects on user experience in VR.

2 RELATED WORK

Thermal display devices have been integrated to VR experiences to improve immersion which enhances user experience. Moreover, rendering thermal properties of virtual objects can be critical to specific applications such as firefighter training [7], where heat perception is needed to learn procedures. Researchers have developed a variety of ways to bring hot and cold sensations to VR users, using air, water, infrared rays and even chemicals, but Peltier elements are undoubtedly the most widely used, as they are inexpensive, relatively easy to control and versatile. An early implementation [1] installed Peltier elements on the armrests of a seated VR system. Amid advances in miniaturisation, Peltier modules have been integrated to multiple wearable devices, allowing localized feedback and higher freedom of movements and interaction in VR. Among these proposals, recent implementations have succeeded in achiev-

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Figure 2: (CAD) StimulHeat integration on the Valve Index controller.

ing very low latency at the cost of heavy instrumentation and a limited runtime [2], or the possibility of catching props but having to carry heavy equipment to transport the power supply [5]. The main difficulty noted by almost every implementations using Peltier elements is obtaining a fast yet lasting sensation, i.e. avoiding overheating of the Peltier when providing a cold feedback. This issue is partly due to ineffective Peltier control, as they are not designed to be controlled by PWM but by DC current variations [6]. Nonetheless, to our knowledge there exists no affordable miniature current controller for Peltier on the market, which has hindered the VR community to make the most of out of Peltier elements' features.

3 IMPLEMENTATION

We present StimulHeat, our small-factor Peltier controller which is efficient and accessible. For demonstration purposes, it is integrated onto a clip-on device for Valve Index controllers that connects via Bluetooth Low Energy (BLE) to Unity based applications.

3.1 Electronics

StimulHeat differs from other existing devices by commanding heat flow rather than the contact temperature between Peltier and skin. Deriving equations of a Peltier element shows that heat transfer depends continuously on the current [4], which means it can be controlled by DC. To this end, we designed a Peltier driver which acts as a controlled invertible current source. The command is a DC current in the range $[-500 \text{ mA}; 500 \text{ mA}]$, differing from PWM control used in other implementations [5]. This type of control mode improves the efficiency of the device, as evidenced by a reduction in consumption [6]. StimulHeat embeds a programmable microcontroller (ref. nRF52840) and its developed firmware is responsible for both wireless communication with a remote host and the control of the Peltier driver by sending current values. These current values are determined by a PID algorithm that receives commands from BLE. Our implementation draws a maximum of 350 mA at a voltage of 3.6 V, which equates to 1.26 W and can be regarded as relatively low compared to other existing prototypes, which exhibit a higher energy consumption in the range of 5 W and above [5] [2].

3.2 Integration Design

StimulHeat is integrated into a Computer-Aided-Design (CAD) model we designed to be attached to the Valve Index controllers via a clip. We chose these specific controllers for their ergonomics and the built-in strap that locks the controller to the hand and keeps the StimulHeat in contact with the skin. Size of CAD model is determined by the Printed Circuit Board (PCB) of the StimulHeat device, which occupies most of the space in front of Peltier element

and wires. It is placed on the controller in a way that avoids affecting its gravity point, ensuring optimal VR user experience. As shown in Fig. 2, the StimulHeat PCB and its integrated battery are placed in the designated space. Three wires emerge from this point and connect to various components via grooves. Two wires are linked to thermistors on both side of the Peltier element to measure temperature and compute the PID heat flow command. Another wire powers the Peltier element, providing the necessary current to control the thermal sensations on the skin's surface.

3.3 Software

Our demo is built with Unity 2022.3 and VR support is handled by the OpenXR package. Interactions and animation of hand poses are managed by the Auto-Hand asset. To implement BLE on both Windows and Android platform and because Unity is not compatible with the native implementation of BLE within the .NET-Core library, we have used third party plugins: BleWinRT¹ and Unity-Android-BLE². To further ease adoption of our device, the delivered Unity API abstracts the hardware layer, enabling developers to use our system on both Windows and Android with only one Unity project. Testing in VR has been conducted on both the Valve Index and Meta Quest 3 Head-Mounted Displays (HMD), for the latter both with Quest Link and autonomously.

4 DEMO WALK-THROUGH

Visitors can experience StimulHeat within an interactive VR scene (see Fig. 1). They are standing and can move within the virtual environment using teleportation. When grabbing a thermally charged virtual objects (e.g., ice or fire material), a flow command is sent to StimulHeat to deliver a congruent thermal feedback.

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¹<https://github.com/adabru/BleWinrtDll>

²<https://github.com/Velorex/Unity-Android-Bluetooth-Low-Energy>